

Green Machines + Metallic materials

Year 9 Science · exam sprint · 2 hours

0:00 Lesson 1 · Green Machines

0:50 Lesson 2 · Metallic materials

1:40 Mini mock (both units)

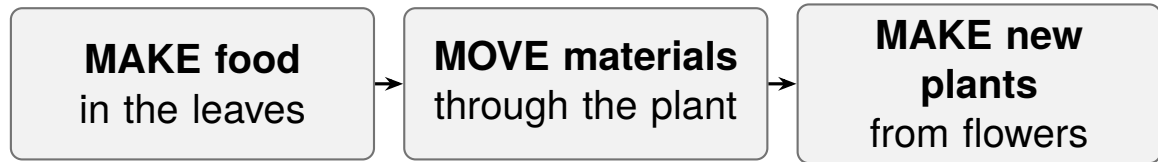
LESSON 1 · Green Machines

Learning intention: explain how a plant makes food, moves water and makes new plants, using the exam words.

Do Now

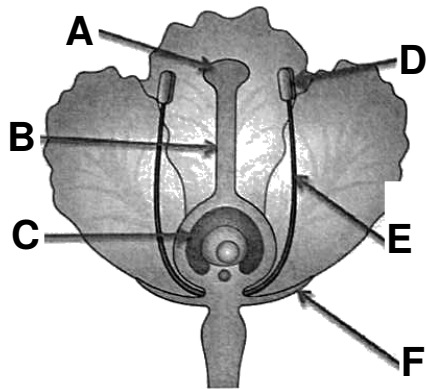
- 1 Which part of a flower makes pollen?
- 2 The pollen nucleus fuses with the ovule nucleus. Is this pollination or fertilisation?
- 3 Name the three conditions a seed needs to germinate.
- 4 Which gas does a leaf take in to make glucose?
- 5 Which tubes carry water up the stem?

The one idea



Every Green Machines question asks about one of these three jobs. Say which job it is, then use that job's words.

Quick review: Parts of a flower



- 1 Name parts A to F.
- 2 Which two parts make up the stamen?
- 3 Which three parts make up the carpel?
- 4 Which part becomes the fruit?

Two ways to make new plants

	Asexual	Sexual
Parents	_____	two
Gametes?	_____	pollen + ovule
Offspring	_____	_____
Examples	runners, bulbs, cuttings	_____

A spider plant grows plantlets on long stems.

Which type of reproduction is this? How do you know?

Why variation matters: write it in this order

- 1 The environment changes, e.g. a new disease arrives.
- 2 Offspring from sexual reproduction are _____ from each other.
- 3 Some offspring have features that let them _____.
- 4 These survive and _____, so the species _____.

A: lines 1–2 **M:** links 2 to 3 **E:** the whole chain + the asexual contrast

Why could a whole field of identical plants die from one disease?

From pollen to seed

Put the cards in order.

- A** The pollen tube grows down the style.
- B** Pollen lands on the stigma.
- C** The ovule becomes a seed; the ovary becomes a fruit.
- D** The pollen nucleus fuses with the ovule nucleus.
- E** The pollen tube enters the ovule through the micropyle.
- F** An insect carries pollen away from an anther.

Which cards are pollination? Which card is fertilisation?

Wind or insect pollination?

Part	Wind-pollinated	Insect-pollinated
Petals	small, dull, no scent	_____
Stigma	_____	sticky, inside the flower
Anthers	hang outside the flower	_____
Pollen	_____	larger, sticky or spiky

Full marks: the feature **and** how it helps.

Why do the anthers of grass hang outside the flower?

Spot the error (1)

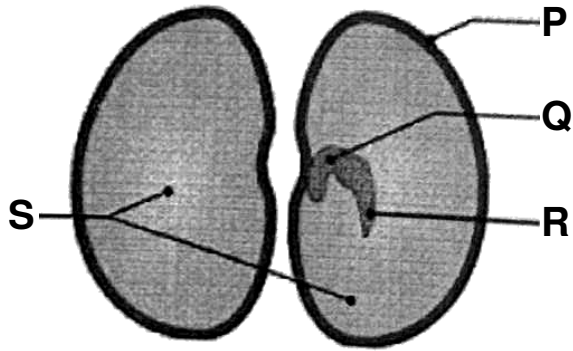
A strawberry plant makes new plants from runners and from its flowers. Compare the two.

Sam's answer

Runners: the new plant gets gametes from the parent, so it is a bit different from it. Flowers: pollen is carried from the stigma to the anther. This is called fertilisation. The seeds show variation because two parents are involved.

Sam's answer has 3 errors. Find each one, fix it, and say why it loses marks.

Quick review: Seeds

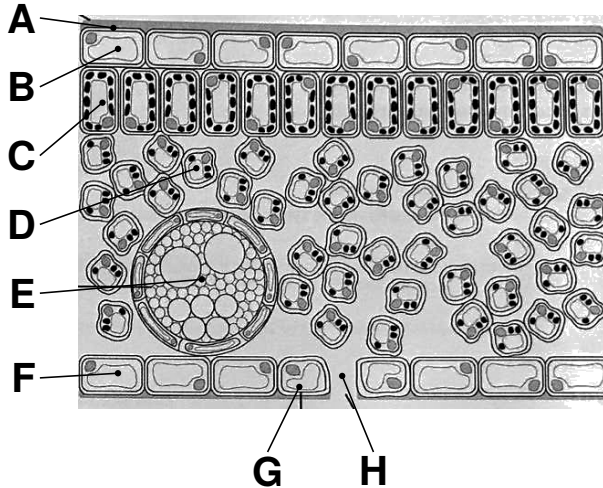


- 1 Name parts P to S.
- 2 Name the three conditions for germination.
- 3 Why are seeds dispersed far from the parent plant?

Practice

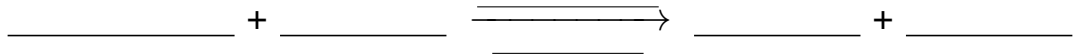
- Q1** A lily has large white petals, a strong scent and nectar. Is it wind- or insect-pollinated? Give two reasons. **[A]**
- Q2** A grower takes cuttings from her best rose bush instead of planting its seeds. Explain why. **[M]**
- Q3** Cress seeds are put in four tubes at 20 °C: A on dry cotton wool; B on wet cotton wool; C on wet cotton wool in a fridge; D under boiled water with a layer of oil on top. Which tubes germinate? Explain the result for D. **[E]**

Quick review: Leaf structure



- 1 Name A to H.
- 2 Which layer does most of the photosynthesis?

Photosynthesis: the word equation



Where does each starting substance enter the plant?

carbon dioxide: through the _____ in the leaf

water: root hairs → _____ → leaf

What does chlorophyll do in this reaction?

Respiration: the same substances, reversed

glucose + _____ $\longrightarrow$ _____ + _____ (+ energy)

	Photosynthesis	Respiration
Day	_____	_____
Night	_____	_____

Which process happens in every living cell, all the time?

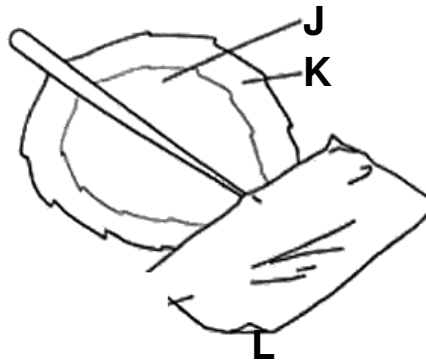
Testing a leaf for starch

Give the reason for each step.

- 1 Put the leaf in boiling water for 1 minute.
- 2 Put it in hot ethanol, standing in a beaker of hot water.
- 3 Rinse it in water.
- 4 Add iodine solution. Blue-black means _____.

Why is the ethanol heated in hot water, not over a Bunsen flame?

Workbook: the foil leaf



A variegated leaf: J green, K white.
Foil L covered part of it for 2 days in the light. Then the leaf was tested with iodine.

Which areas turn blue-black? Explain each area.

Spot the error (2)

Write the word equation for photosynthesis. Then say how you would show that a leaf has made food.

Sam's answer

oxygen + water makes glucose + carbon dioxide (light above the arrow). Boil the leaf, put it in hot ethanol, rinse it, then add iodine. If it goes blue-black, the leaf has made glucose. Plants photosynthesise in the day and respire at night.

Sam's answer has 3 errors. Find each one, fix it, and say why it loses marks.

Practice

- Q1** Two potted plants are kept in the light for 2 days, each under a sealed bell jar. Jar 1 also holds a dish of soda lime, which absorbs carbon dioxide. Jar 2 holds a dish of water. A leaf from each plant is tested with iodine. Predict and explain the results. **[M]**
- Q2** In a greenhouse full of plants, the carbon dioxide level rises at night and falls during the day. Explain why. **[E]**

Quick review: Transport words

Match each word to its job.

- | | |
|----------------------|---|
| 1 xylem | A carries glucose from the leaves |
| 2 phloem | B lets gases and water vapour in and out of the leaf |
| 3 root hair | C absorbs water from the soil |
| 4 stomata | D carries water and minerals up the plant |
| 5 guard cells | E change shape to open and close the stomata |

How water travels through a plant

soil water

↓ by _____

root hair cell

↓

_____ in the root and up the stem

↓

leaf cells: water evaporates

↓

out through the _____ as water vapour

What is the loss of water vapour from the leaves called?

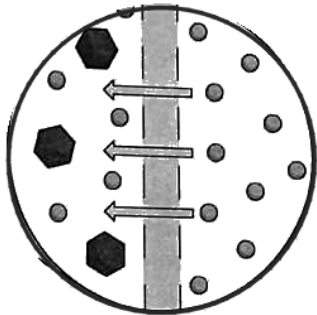
Why the water moves up: write it in this order

A celery stalk stands in blue dye for 2 days. The leaves turn blue.

- 1 Water _____ from the leaves through the stomata.
- 2 This _____ more water up the _____.
- 3 The dye travels with the water, so the _____ turn blue.
- 4 Cut the stalk: the blue dots show where the _____ are.

A: dye goes up **M:** names xylem **E:** transpiration causes the pull

Osmosis

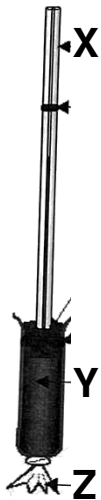


Left: sugar solution. Right: water.

- 1 Which shape is a water molecule?
- 2 Which way do the water molecules move?
- 3 Why can the sugar molecules not cross?

Osmosis is the movement of _____ from a _____ concentration of water to a _____ concentration of water through a _____ membrane.

Workbook: the Visking tubing model



The tubing Z sits in a beaker of water. Over 30 minutes the liquid rises up X.

- 1 Which part acts like the root hair? Which acts like the stem?
- 2 Explain why the liquid rises.
- 3 A root hair is long and thin. How does this help it?

Spot the error (3)

A white carnation stands in blue dye. A day later its petals are blue. Explain.

Sam's answer

The blue water was carried up the phloem tubes. The flower sucks the water up because it is thirsty. Water always gets into a plant by osmosis, from a low concentration of water to a high concentration of water.

Sam's answer has 3 errors. Find each one, fix it, and say why it loses marks.

Practice

Q1 Name the process that moves water into a root hair, and the tubes that carry it to the leaves. [A]

Q2 In a Visking tubing model the liquid is 12 mm up the tube at the start, 30 mm after 10 minutes and 48 mm after 20 minutes.

(a) Calculate the average rate of rise in mm per minute.

(b) Explain why the liquid rises. [M]

Q3 A plant wilts when its roots are put in very salty water. Explain why. [E]

Ways to lose marks: Green Machines

- L1** Pollination moves pollen; fertilisation fuses the nuclei.
- L2** Asexual: one parent, no gametes, identical offspring.
- L3** Feature **and** how it helps.
- L4** Light and chlorophyll go above and below the arrow.
- L6** Iodine shows starch, not glucose.
- L7** Xylem carries water; phloem carries glucose.

Summary: Green Machines

Flowers: pollination moves pollen; fertilisation joins the nuclei.

Seeds need water, warmth and oxygen; dispersal cuts competition.

Leaves: carbon dioxide + water make glucose + oxygen, using light.

Water: osmosis into root hairs, up the xylem, out by transpiration.

Classwork: Green Machines

On your classwork sheet, about 10 minutes:

- A** Flowers and seeds
- B** Photosynthesis and respiration
- C** Transport and osmosis, with a graph to draw

Write in full sentences for every **explain** question.

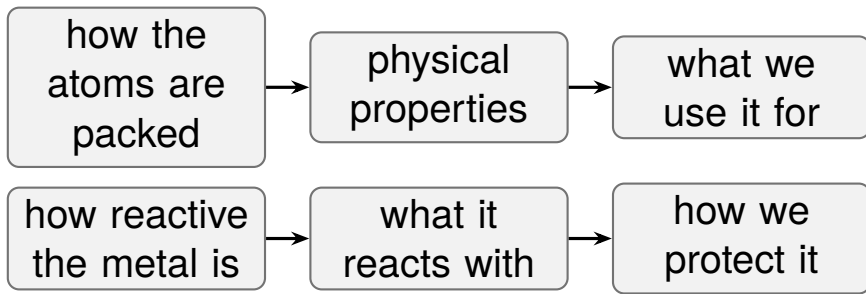
LESSON 2 · Metallic materials

Learning intention: link properties to uses, calculate density, and explain alloys, rusting and the reactions of metals.

Do Now

- 1 Give two physical properties shared by most metals.
- 2 Name the three common magnetic metals.
- 3 Write the formula for density.
- 4 Gallium melts at 30°C . Is it solid or liquid on a 35°C day?
- 5 A metal is heated red hot, then cooled slowly. Name this heat treatment.

The one idea



Top row: physical properties. Bottom row: chemical reactions.
Every Metals question sits on one of these two rows.

Quick review: Property words

Match each word to its meaning, then spell it.

1 malleable

A has a shiny surface that reflects light

2 ductile

B can be hammered into shape

3 lustrous

C makes a ringing sound when hit

4 sonorous

D can be stretched into wires

5 conductor

E lets heat or electric current pass through

Choosing a metal

Metal	Melting point (°C)	Strength	Electrical conductivity	Density (g/cm³)
W	1450	high	low	7.8
X	650	low	high	2.7
Y	1080	medium	very high	8.9
Z	230	low	medium	7.3

Which metal for (a) a bridge frame, (b) wiring in a house, (c) a light overhead power cable? (d) Which would melt in a Bunsen flame (about 800 °C)?

Justify a choice: write it in this order

- 1 Name the metal.
- 2 Give the property that matters for this use.
- 3 Say why that property matters for the use.
- 4 Compare: why not another metal? Use the data.

A: 1 + 2 **M:** 1 to 3 **E:** 1 to 4 with numbers

For the power cable I choose _____ because it _____.

This matters because _____.

Y conducts better, but _____.

Density: the formula and the volume

density = $\frac{\text{mass}}{\text{volume}}$

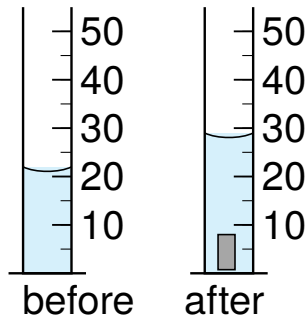
units: g/cm^3 or kg/m^3

Volume of a regular block:

Volume of an odd-shaped object:

the _____ method

What is the volume of the object on the right?



Density: worked example

A brass key has a mass of 25.5 g. When it is lowered into a measuring cylinder, the water level rises from 30.0 mL to 33.0 mL.

Step 1 volume = _____ - _____ = _____ mL

Step 2 density = _____ $\div$ _____

Step 3 density = _____ with its unit

Check: is your answer bigger than 1 g/mL? Brass sinks, so it should be.

Spot the error (4)

A metal bolt has a mass of 39.5 g. When it is put into a measuring cylinder, the water rises from 20.0 mL to 25.0 mL. Calculate the density of the bolt.

Sam's answer

volume = 25.0 mL

density = $39.5 \times 25.0 = 987.5 \text{ g}$

Sam's answer has 3 errors. Find each one, fix it, and say why it loses marks.

Practice

- Q1** A metal block is $2.0\text{ cm} \times 2.0\text{ cm} \times 3.0\text{ cm}$ and has a mass of 64.8 g . Calculate its density. **[A]**
- Q2** A ring has a mass of 31.5 g . It makes the water in a measuring cylinder rise from 40.0 mL to 43.0 mL . Which metal is it? **[M]**
iron 7.9 copper 8.9 silver 10.5 aluminium 2.7 (g/mL)
- Q3** Silver conducts electricity better than copper. Explain why house wiring is made of copper. **[E]**

Quick review: Heat treatments

- 1 Heat red hot, then cool quickly in cold water.
- 2 Heat red hot, then cool slowly.
- 3 Heat strongly until a blue sheen appears.

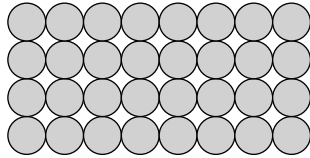
Name each treatment: annealing, quenching or tempering.

Then match it to a result: **A** soft, easy to bend **B** hard but brittle
C tough and springy

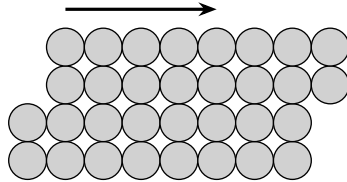
- 4 A hair clip must spring back into shape. Which treatment does it need?

Why alloys are harder (1)

A pure metal: every atom is the same size, in regular layers.



before

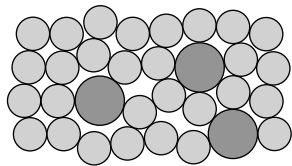


after a push

What did the layers do when they were pushed? So is a pure metal hard or soft?

Why alloys are harder (2)

An alloy: a metal mixed with another element.



- 1 The atoms are _____ sizes.
- 2 This _____ the regular layers.
- 3 The layers cannot _____ easily.
- 4 So the alloy is _____ than the pure metal.

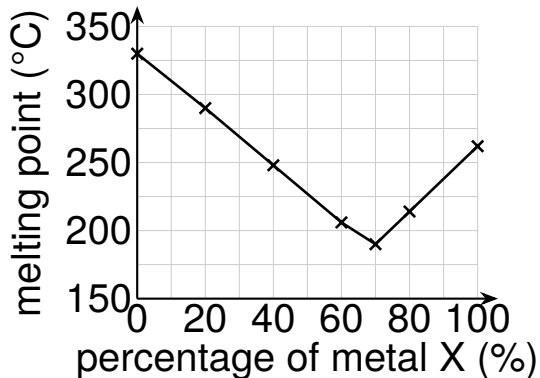
A: harder **M:** + different-sized atoms **E:** whole chain

Alloys in use

Alloy	Main metals	Use
bronze	copper + _____	gears, machine bearings
solder	tin + lead	_____
stainless steel	iron, chromium, nickel	_____
nitinol	nickel + titanium	_____

Why is solder good for joining electrical wires?
Is an alloy a compound or a mixture?

Melting points of alloys



Mixtures of two metals, X and Y.

- 1 What is the melting point of pure X? Of pure Y?
- 2 Which mixture has the lowest melting point?
- 3 Describe the trend from 0% to 70% X.

Practice

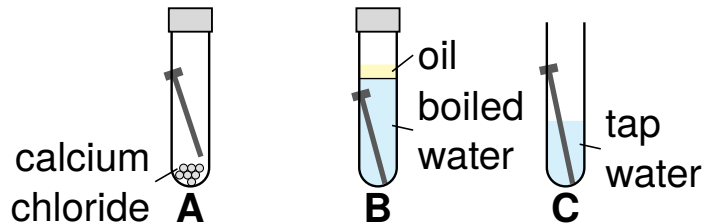
- Q1** Name the heat treatment that makes steel hard but brittle. **[A]**
- Q2** Brass is an alloy of copper and zinc. Explain why brass is harder than pure copper. **[M]**
- Q3** A car spring and a car door panel are both made of steel. Explain which heat treatment each one should have. **[E]**

Quick review: Corrosion and rusting

True or false?

- 1 Rusting is a type of corrosion.
- 2 Any metal can rust.
- 3 Salt water makes iron rust faster.
- 4 Gold corrodes quickly in air.
- 5 Corrosion is a chemical reaction of a metal with substances such as oxygen and water.

Rusty nail experiment (1): the set-up



Iron nails, left for 3 days.

A The calcium chloride removes _____.

B Boiling removes _____ from the water. The oil stops _____.

C The nail has both _____ and _____.

Rusty nail experiment (2): write it in this order

Results after 3 days: A _____ B _____ C _____

- 1 Only tube _____ rusted.
- 2 Tube A had air but no _____.
- 3 Tube B had water but no _____.
- 4 So iron needs _____ to rust.

A: C rusted **M:** + what A and B lacked **E:** the conclusion from all three

Spot the error (5)

Steel paper clips are sealed in three tubes: P with calcium chloride (dry air), Q in boiled water under a layer of oil, R in tap water with air above. Describe and explain the results.

Sam's answer

P: no rust, because there is no air. Q: a little rust, because it has water. R: lots of rust. So iron only needs water to rust.

Sam's answer has 3 errors. Find each one, fix it, and say why it loses marks.

Preventing rust

Object	Method	Why this method
steel fence	_____	large, cheap, easy to redo
door hinge	oil or grease	_____
rubbish bin	_____	zinc coating
kitchen sink	_____	never needs a coating

Every method keeps _____ and _____ away from the iron.

Practice

Q1 Name the two substances iron needs in order to rust. **[A]**

Q2 Explain why a car in a seaside town rusts faster than the same car far inland. **[M]**

Q3 Plan an experiment to find out whether salt water makes iron nails rust faster than tap water. Say what you would change, what you would measure, and what you would keep the same. **[E]**

Quick review: Word equation patterns

- 1 metal + oxygen $\rightarrow$ _____
- 2 metal + water $\rightarrow$ _____ + _____
- 3 metal + hydrochloric acid $\rightarrow$ _____ + _____
- 4 Test for hydrogen: a lit splint gives a _____.

Writing word equations

(a) Iron wool burns in oxygen.

iron + oxygen $\rightarrow$ _____

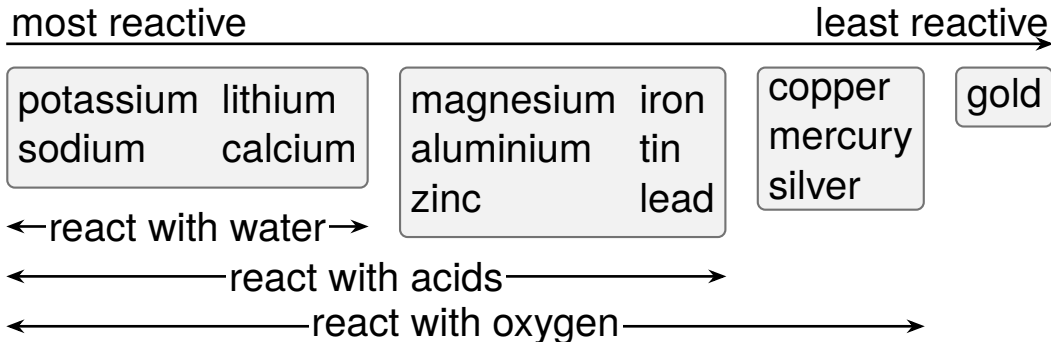
(b) Lithium fizzes in cold water.

lithium + water $\rightarrow$ _____ + _____

(c) Aluminium powder in dilute hydrochloric acid.

(d) Gold in dilute hydrochloric acid.

The reactivity series



- 1 Would tin react with dilute acid?
- 2 Why is gold found in the ground as pure metal, but iron is not?

Spot the error (6)

Complete each word equation, or write “no reaction”.

Sam's answer

1. calcium + oxygen makes calcium oxygen
2. iron + hydrochloric acid makes iron chloride + water
3. potassium + water makes potassium hydroxide + hydrogen
4. silver + hydrochloric acid makes silver chloride + hydrogen

Sam's answer has 3 errors. Find each one, fix it, and say why it loses marks.

Practice

Q1 Write the word equation for aluminium reacting with oxygen. **[A]**

Q2 Four metals are added to dilute acid. Put them in order of reactivity, most reactive first, and give your evidence. **[M]**

P many bubbles, quickly

Q no bubbles

R a few bubbles, slowly

S reacts even with cold water

Q3 Describe an experiment with dilute hydrochloric acid to show that zinc is more reactive than copper. Give the result you expect. **[E]**

Ways to lose marks: Metals

L10 Density = mass $\div$ volume, unit g/mL or g/cm³.

L11 Volume = final reading – first reading.

L12 Property + why it matters + comparison.

L13 Different-sized atoms, layers cannot slide.

L14 Rusting needs water **and** oxygen.

L15 Oxide; hydroxide + hydrogen; chloride + hydrogen.

Summary: Metallic materials

Properties decide uses: say the property, why it matters, and compare.

Density = mass $\div$ volume; displacement gives the volume.

Alloys are harder: different-sized atoms stop the layers sliding.

Rust needs water and oxygen; reactivity decides how metals react.

Classwork: Metallic materials

On your classwork sheet, about 10 minutes:

- D** Properties and density
- E** Alloys and heat treatment
- F** Corrosion and reactions, with an experiment to plan

Show every step of a calculation, with units.

Mini mock

17 minutes · both units · on the paper

Answer every part.

Each question is marked A, M or E.

Explain questions: write it in the order you practised today.

Homework

hw1 Green Machines, about 40 minutes

hw2 Metallic materials, about 40 minutes

Also: Metals booklet p.10–11, draw the solder graph and answer Q1–2.

Do them on two different days before the exam.